BREADTH FIRST SEARCH (BFS)

Data Structures / C Programming

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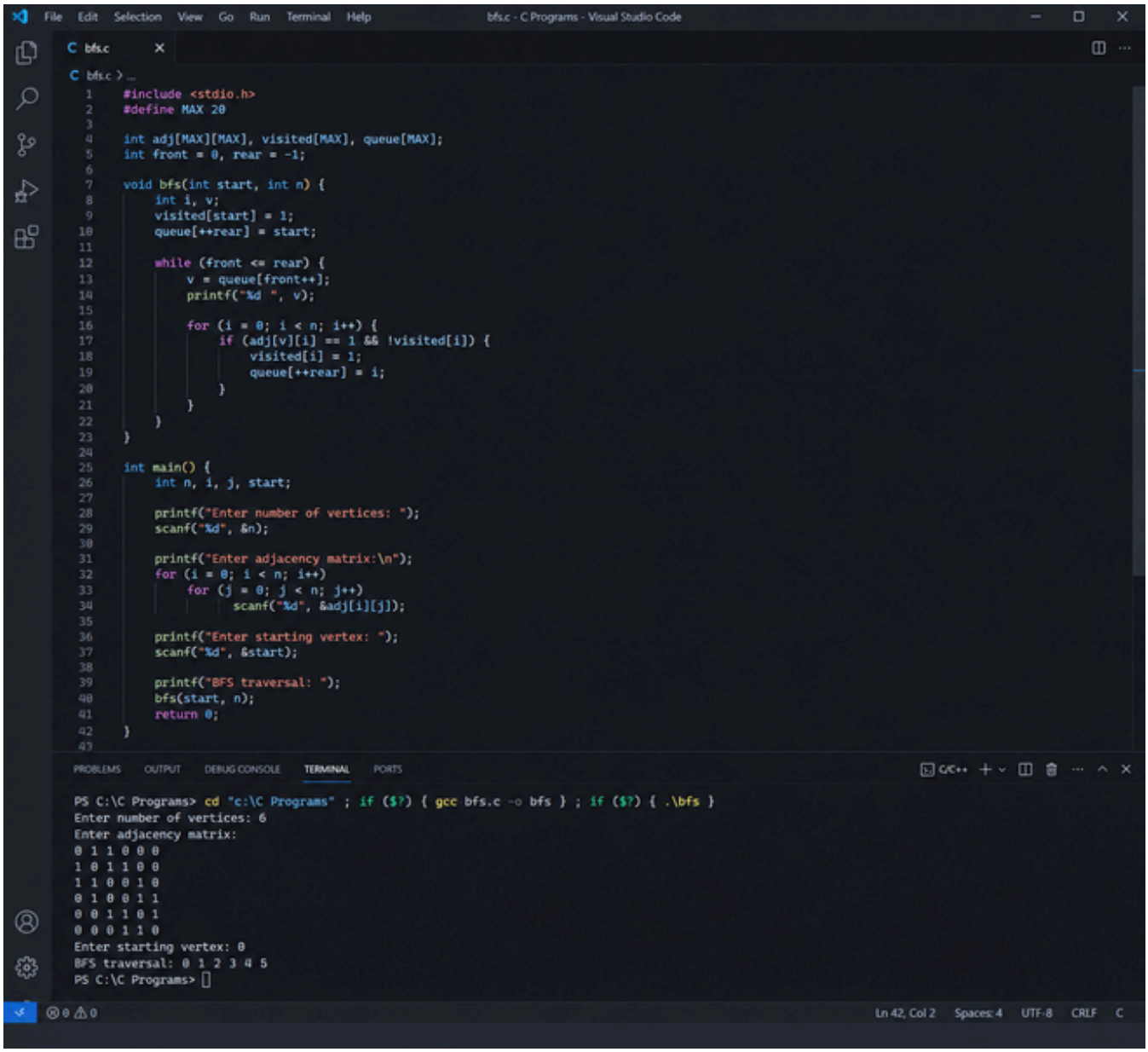
Aim

To implement Breadth First Search (BFS) traversal of a graph using an adjacency matrix and a queue in the C programming language.

Program

```
#include <stdio.h>#define MAX 20int adj[MAX][MAX], visited[MAX], queue[MAX];int front = 0, rear = -1;void bfs(int start, int n) {    int i, v;    visited[start] = 1;    queue[++rear] = start;    while (front <= rear) {        v = queue[front++];        printf("%d ", v);        for (i = 0; i < n; i++) {            if (adj[v][i] == 1 && !visited[i]) {                visited[i] = 1;                queue[++rear] = i;            }        }    }}int main() {    int n, i, j, start;    printf("Enter number of vertices: ");    scanf("%d", &n);    printf("Enter adjacency matrix:\n");    for (i = 0; i < n; i++)        for (j = 0; j < n; j++)            scanf("%d", &adj[i][j]);    printf("Enter starting vertex: ");    scanf("%d", &start);    printf("BFS traversal: ");    bfs(start, n);    return 0;}
```

VS Code Program and Output



Conclusion

The Breadth First Search algorithm was successfully implemented in C using a queue. The graph was traversed level by level from the selected starting vertex, and the BFS traversal was displayed successfully.